

Kuan-Ting (Larry) Chen

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Education

National Yang Ming Chiao Tung University, [Advanced Database System Laboratory](#)

Hsinchu, Taiwan

M.S. in Computer Science and Engineering (Advisor: Prof. Wen-Chih Peng)

2024/09 – Present

- Research: Sports Analytics, Large Language Models, and Reinforcement Learning
- GPA: 4.3/4.3; Member, The **Phi Tau Phi** Scholastic Honor Society; 2 Academic Excellence Awards; 2 publications

National Cheng Kung University, [Intelligent Knowledge Management Laboratory](#)

Tainan, Taiwan

B.S. in Computer Science and Information Engineering (Advisor: Prof. Hung-Yu Kao)

2020/09 – 2024/06

- Research: Natural Language Processing, Hateful Meme Detection, and Fact Checking
- GPA: 4.21/4.3; 1 Academic Excellence Award; 3 extracurricular awards

Experience

Student Project Lead, [Precision Sports Science Research Project](#)

2025/05 – 2026/09

- Coordinated five subprojects as the student lead, managing cross-team integration, reports, meetings, and project demonstrations
- Developed CoachAI+ badminton environment and BadmintonTwin, and contributed to BadmintonReels and BadmintonGPT

Projects

BadmintonTwin: An LLM-based Virtual Badminton Player

- Developed a multi-agent LLM framework for controllable, player-specific badminton rally simulation
- Built a hybrid RAG pipeline combining physical-state retrieval and semantic contextual memory from historical rallies

Skills: Large Language Models · Retrieval-Augmented Generation · AI Agents · Human Evaluation

CoachAI+ Badminton Environment

[Code](#)

- Developed PDQN, HybridPPO, and HybridSAC baselines for hybrid action-space RL in badminton strategy optimization
- Designed simulation environments and evaluation metrics by integrating coach knowledge into quantifiable objectives

Skills: Reinforcement Learning · Hybrid Action Space · Simulator · PyTorch · Sport Analytics

Publications

Kuan-Ting Chen, Kuang-Da Wang, and Wen-Chih Peng. "BadmintonTwin: Tactical Memory-Augmented LLM Agents for Instruction-Conditioned Badminton Rally Simulation." **Under review at ACL Rolling Review.**

Tsan-Tsung Yang, I-Wei Chen, Kuan-Ting Chen, Shang-Hsuan Chiang, and Wen-Chih Peng. "Robust Detection and Source Identification of AI-Generated Images Using CNN and CLIP-Based Models." **AAAI Defactify 2025.** [\[Paper\]](#) [\[Code\]](#)

Awards

College Student Research Creativity Award

2024/07

- Awarded to only ~5% of participants. Converted multimodal inputs and HurtLex knowledge into text, then finetuned HateBERT with Masked Language Modeling (MLM) objective function.

Skills: Hateful Meme Detection · Visual Feature Extraction · Prompt Engineering · MLM · CLIP

AI Cup 2023 Factual Text Retrieval and Verification Competition – Honorable Mention

2024/03

- Achieved the 13th place out of 301 participants. Built a 3-stage fact verification pipeline using multi-source retrieval, BERT-based evidence selection, and BERT-based claim verification.

Skills: BERT · Fact Verification · Information Retrieval · TF-IDF · Chinese Text Preprocessing · Text Classification

Meichu Hackathon Hacker Group (TSMC) – Second Place

2022/10

- Trained and optimized YOLOv7 with augmentation and multi-stage training, then deployed real-time inference on laptops and Raspberry Pi with REST API integration.

Skills: YOLOv7 · Data Augmentation · Object Detection · Computer Vision · Raspberry Pi

Technical Skills

Proficient: Python, PyTorch, AI Agents, RAG, Reinforcement Learning, Computer Vision

Familiar: C++, JavaScript

Tools: Git, VS Code, OpenCode, vLLM

Languages: Mandarin (Native), English (Fluent; TOEIC: 800)